

Semantic-Aware Cross-Layer Resource Allocation in Molecularly Attenuated THz Channels for 6G Swarm Architectures

Fernando May Fuentes
Instituto Politécnico Nacional
Ciudad de México, México
fmayf1500@alumno.ipn.mx

ABSTRACT

Terahertz (THz) communications offer large bandwidth but suffer from severe path loss and molecular absorption. This paper presents a reproducible first-order semantic-encoding artifact for studying the interaction between a THz channel model and latent representations. The current implementation is a deterministic autoencoder-style model, not a variational autoencoder, and it does not yet implement a valid rate–distortion comparison against raw transmission. Across one synthetic run, semantic fidelity ranged from 0.656 to 0.670 over 10–1000 m, while the modeled SNR decreased from -14.4 to -374.4 dB. These results establish a baseline and expose the need for calibrated channel parameters and repeated trials before making efficiency claims.

KEYWORDS

semantic communication, THz, 6G, VAE, molecular absorption, swarm architecture

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1 INTRODUCTION

THz communications (0.1-10 THz) promise terabit-per-second data rates for 6G networks. However, molecular absorption—particularly from water vapor—creates frequency-dependent attenuation that severely limits practical range. Rather than transmitting every bit with equal priority, semantic communication transmits only the meaning relevant to the task.

The scope of this paper is deliberately limited to an artifact-level baseline. We do not claim a validated semantic communication system, a calibrated THz channel, or a measured throughput gain. The objective is to expose the assumptions required for a defensible cross-layer experiment and provide a reproducible starting point for later work.

ORCID: 0009-0002-3953-5224.

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2 RELATED WORK

Task-oriented and semantic communication systems optimize task utility rather than bit-perfect reconstruction. Learned joint source-channel coding can improve robustness under constrained channels, but comparisons require a task metric and a matched physical-layer budget. THz channel models additionally require frequency-selective molecular absorption and realistic link budgets. Our artifact currently implements only a simplified analytical attenuation model and therefore occupies the baseline, not the final-system, position in this literature.

3 SYSTEM MODEL

3.1 THz Channel Model

Molecular absorption coefficient:

$$K(f) = K_{\text{peak}} \cdot \exp\left(-\frac{(f - f_0)^2}{2\sigma^2}\right) \quad (1)$$

Signal attenuation:

$$P_r = P_t \cdot \exp(-K(f) \cdot d) \cdot \left(\frac{\lambda}{4\pi d}\right)^2 \quad (2)$$

3.2 Semantic Fidelity

$$\mathcal{F}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{1}{1 + \|\mathbf{x} - \hat{\mathbf{x}}\|^2} \quad (3)$$

3.3 Encoder and Resource Allocation

The implementation maps an input vector $\mathbf{x} \in \mathbb{R}^{128}$ to a 16-dimensional latent vector using a tanh projection. Subchannel weights are proportional to the mean absolute latent activation. This is a deterministic autoencoder-style allocation rule; it is not a probabilistic VAE and it does not perform channel-aware power control.

4 EXPERIMENTAL PROTOCOL

The experiment uses carrier frequency 0.3 THz, bandwidth 10 GHz, temperature 300 K, relative humidity 0.5, and distances of 10, 50, 100, 200, 500, and 1000 m. At each distance, eight synthetic samples of dimension 128 are generated using NumPy seed 20260909. We report exact single-run values and do not compute confidence intervals from this one realization. The comparison routine returns zero rates because the modeled SNR is negative at all evaluated distances; therefore no bandwidth-efficiency claim is made.

5 RESULTS

Semantic fidelity remained in the range 0.656–0.685 in this run, while modeled SNR degraded by approximately 360 dB. Because the current decoder does not inject channel noise into the latent

Table 1: THz Channel Performance at Various Distances

Distance	SNR (dB)	Semantic Fidelity
10m	-14.4	0.672
50m	-41.3	0.667
100m	-63.5	0.664
200m	-101.8	0.673
500m	-206.8	0.656
1000m	-374.4	0.685

representation, this stability must not be interpreted as communication robustness. It is a baseline reconstruction result only.

6 LIMITATIONS AND THREATS TO VALIDITY

The channel model is a simplified analytic approximation, the humidity and absorption parameters are not calibrated against a standard channel database, and the autoencoder is not a VAE. The artifact also lacks a raw-bit baseline, semantic task metric, coding/modulation

layer, and confidence intervals. Those components are required before submission as a communications result.

7 CONCLUSION AND REPRODUCIBILITY

This work establishes a transparent baseline rather than a completed THz semantic link. A valid next experiment must add channel noise, a matched raw-bit baseline, a task-level semantic metric, repeated seeds, and calibrated absorption data before any claim of gain is made.

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